

HIRDESH SINGH

+91 9084130259 | 22hirdesh@gmail.com | [LinkedIn](#) | [GitHub](#)

PROFESSIONAL SUMMARY

Computer Science undergraduate with a strong foundation in Data Structures & Algorithms, Object-Oriented Programming, DBMS, and Operating Systems. Experienced in full-stack software development, building backend systems, REST APIs, and AI-driven applications using Java, C++, JavaScript, Node.js, React.js, and MongoDB. Skilled in database design, authentication workflows, debugging, and scalable software engineering practices.

TECHNICAL SKILLS

Programming Languages: Java, C++, JavaScript, TypeScript, Python, SQL

Core Computer Science: Data Structures & Algorithms, Object-Oriented Programming, DBMS, Operating Systems, SDLC

Backend & APIs: Node.js, Express.js, REST API Design, MVC Architecture, Spring Boot/MVC (Basic)

Frontend: React.js, HTML5, CSS3, Bootstrap

Databases: MongoDB, MySQL, PostgreSQL (working knowledge)

AI / Computer Vision: CNNs, RNNs, LSTMs, YOLOv8, OpenCV

Developer Tools: Git, GitHub, Postman, VS Code, Agile/Scrum, CI/CD (familiar), Debugging & Testing

PROJECTS

Medicare — Online Disease Prediction & Doctor Appointment System

HTML5, CSS3, JavaScript, MongoDB, REST APIs | Live: doctor-meds-frontend.onrender.com

- Engineered a responsive, multi-role (Admin, Doctor, User) full-stack interface supporting appointment booking and disease-prediction workflows.
- Designed MongoDB database schemas and implemented CRUD operations for users, doctors, and appointments.
- Built REST APIs for authentication, scheduling, and role-based access control.
- Applied OOP principles and modular MVC-style design to improve maintainability and debugging efficiency.

NextGen Ballot — Online Voting Platform

Node.js, Express.js, MongoDB Atlas, Mongoose, JavaScript | Live: nextgen-ballot.onrender.com

- Developed a secure full-stack voting application using Node.js, Express.js, MongoDB Atlas, and Mongoose.
- Implemented OTP authentication, CAPTCHA verification, Aadhaar number format validation, and session-based login for secure access.
- Designed and tested REST APIs for voter registration, vote casting, result retrieval, and admin operations.
- Deployed the application on Render and managed version control and collaboration using Git and GitHub.

Smart Navigation System for Visually Impaired

Python, YOLOv8, OpenCV, Flask, pyttsx3

- Built a real-time object and obstacle detection system using YOLOv8 and OpenCV on live camera input.
- Structured a modular pipeline (detection → decision logic → voice alerts) integrated with a Flask backend and web interface.
- Collaborated in an Agile workflow, iterating across sprints to integrate the computer-vision pipeline with the backend and UI.

EDUCATION

Bachelor of Technology in Computer Science and Engineering

VIT Bhopal University | CGPA: 8.47/10 | Expected 2027

ACHIEVEMENTS

- Solved 250+ Data Structures and Algorithms problems across LeetCode and GeeksforGeeks, strengthening algorithmic problem-solving skills.
- Participated in multiple hackathons, building scalable full-stack applications under time constraints.

CERTIFICATIONS

- Generative AI Professional — Oracle
- Oracle Certified Foundations Associate — Oracle
- Google IT Support Professional Certificate — Google
- Bits and Bytes of Computer Networking — Coursera